# Solutions to Linear Algebra Problems

Minh Le

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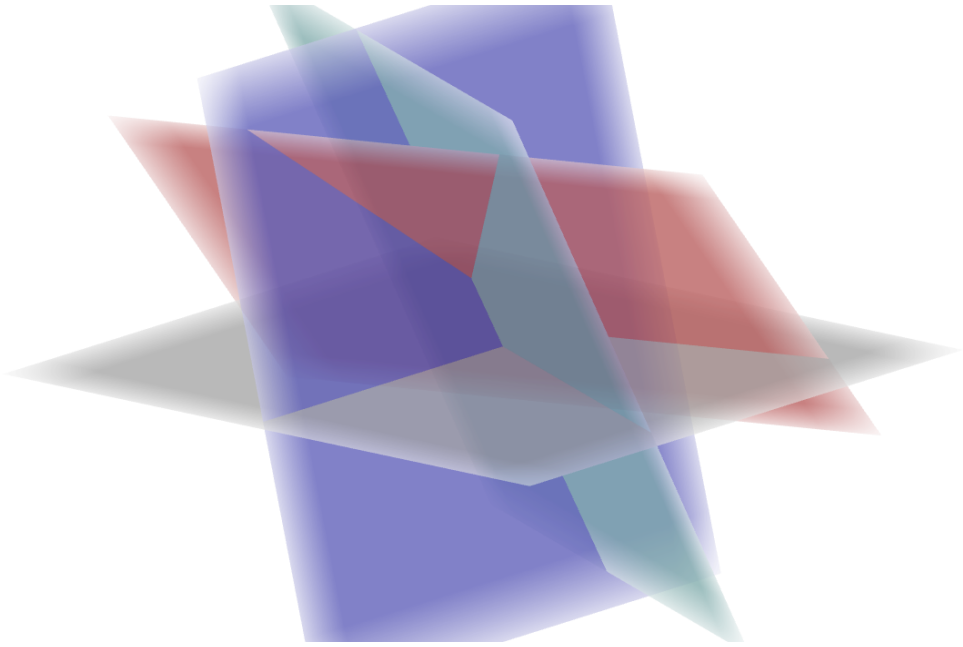
**Abstract**

This note collects my solutions to selected problems in Linear Algebra. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- Linear Algebra Done Right 4e, Sheldon Axler



**Problem 1.**

Prove that the union of two subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces is contained in the other.

*Source: Linear Algebra Done Right - Ex.12 p.25*

***Solution:***

Let $V_1, V_2$ be two subspaces of $V$.
($\Rightarrow$): Suppose that $V_1 \cup V_2$ is a subspace of $V$. Suppose for contradiction that one of the subspaces $V_1$ or $V_2$ is not contained in the other. We can choose $u, w$ such that $u \in V_1, u \notin V_2$ and $w \in V_2, w \notin V_1$.
Since $V_1 \cup V_2$ is a subspace, we have $u + w \in V_1 \cup V_2$. This means $u + w$ must be in $V_1$ or $V_2$.
If $u + w \in V_1$, since $V_1$ is a subspace and $-u \in V_1$, then $w = (u + w) - u \in V_1$, which is a contradiction. Similarly, if $u+w \in V_2$, since $V_2$ is a subspace and $-w \in V_2$, then $u = (u+w)-w \in V_2$, which is also a contradiction.
Therefore, $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.
($\Leftarrow$): If $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$, then $V_1 \cup V_2 = V_2$ or $V_1 \cup V_2 = V_1$, which is a subspace.

**Problem 2.**

Suppose $v_1, \ldots, v_m$ is a list of vectors in $V$. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list $v_1, \ldots, v_m$ is linearly independent if and only if the list $w_1, \ldots, w_m$ is linearly independent.

*Source: Linear Algebra Done Right - Ex.14 p.38*

***Solution:***

($\Rightarrow$): Suppose the list $v_1, \ldots, v_m$ is linearly independent. Consider a combination of $w_1, \ldots, w_m$ such that:

$$\begin{aligned} c_1w_1 + c_2w_2 + \cdots + c_mw_m &= 0 \\ \Rightarrow c_1v_1 + c_2(v_1 + v_2) + \cdots + c_m(v_1 + v_2 + \cdots + v_m) &= 0 \\ \Rightarrow (c_1 + c_2 + \cdots + c_m)v_1 + (c_2 + \cdots + c_m)v_2 + \cdots + c_mv_m &= 0 \end{aligned}$$

Since $v_1, \ldots, v_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $w_1, \ldots, w_m$ are linearly independent.
($\Leftarrow$): Suppose $w_1, \ldots, w_m$ are linearly independent. Consider a combination of $v_1, \ldots, v_m$ such

that:

$$\begin{aligned} c_1v_1 + c_2v_2 + \cdots + c_mv_m &= 0 \\ \Rightarrow c_1w_1 + c_2(w_2 - w_1) + c_3(w_3 - w_2) + \cdots + c_m(w_m - w_{m-1}) &= 0 \\ \Rightarrow (c_1 - c_2)w_1 + (c_2 - c_3)w_2 + \cdots + (c_{m-1} - c_m)w_{m-1} + c_mw_m &= 0 \end{aligned}$$

Since $w_1, \ldots, w_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $v_1, \ldots, v_m$ are linearly independent.

**Problem 3.**

Suppose $V$ is finite-dimensional and $U, W$ are subspaces of $V$ such that $V = U + W$. Prove that there exists a basis of $V$ consisting of vectors in $U \cup W$.

*Source: Linear Algebra Done Right - Ex.5 p.43*

***Solution:***

Let $u_1, u_2, \ldots, u_m$ be a basis of $U$ and $w_1, w_2, \ldots, w_n$ be a basis of $W$.
Since $V = U + W$, we have the list $\{u_1, u_2, \ldots, u_m, w_1, w_2, \ldots, w_n\}$ spans $V$.
Applying the procedure in Theorem 2.30 to reduce this list to a basis of $V$ produces a basis consisting of the vectors $u_1, \ldots, u_m$ and some of the $w$'s (none of the $u$'s get deleted in this procedure because $u_1, \ldots, u_m$ is linearly independent).

**Problem 4.**

Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$ with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of $V$, each of dimension $n - 1$, whose intersection equals $U$.

*Source: Linear Algebra Done Right - Ex.16 p.49*

***Solution:***

Let $u_1, u_2, \ldots, u_m$ be a basis of $U$. This list can be extended to a basis of $V$ by adjoining $n - m$ additional vectors (by Theorem 2.32), say $w_1, w_2, \ldots, w_{n-m}$. We denote this extended list as $L$.
Let $V_i$ $(1 \leq i \leq n-m)$ be the subspace spanned by $L$ without including $w_i$. Clearly $\dim V_i = n-1$ and $\bigcap_{i=1}^{n-m} V_i$ is also a subspace that contains $U$.

Consider $v \in \bigcap_{i=1}^{n-m} V_i$. We have:

$$\begin{aligned} a_1u_1 + a_2u_2 + \cdots + a_mu_m + b_2w_2 + b_3w_3 + \ldots \\ +b_{n-m}w_{n-m} = v \quad (\text{since } v \in V_1) \end{aligned}$$

$$\begin{aligned} c_1u_1 + c_2u_2 + \cdots + c_mu_m + d_1w_1 + d_3w_3 + d_4w_4 \\ + \cdots + d_{n-m}w_{n-m} = v \quad (\text{since } v \in V_2) \end{aligned}$$

By subtracting, we have:

$$\begin{aligned} (a_1 - c_1)u_1 + (a_2 - c_2)u_2 + \cdots + (a_m - c_m)u_m - d_1w_1 \\ + (b_2 - d_2)w_2 + b_3w_3 + \cdots + (b_{n-m} - d_{n-m})w_{n-m} = 0 \end{aligned}$$

Since $L$ is a list of independent vectors and by repeating this process, this implies that all coefficients for $w_i$'s are 0 and $v$ is uniquely represented in the basis of $U$.
We conclude that $\bigcap_{i=1}^{n-m} V_i = U$.